

Chinyere Gloria Mbaka

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EDUCATION

Rochester Institute of Technology

Aug 2023 – Present

Ph.D. Student, Electrical and Computer Engineering; GPA: 3.62/4.00

Rochester, NY

Selected coursework: Real-Time and Embedded Systems; Applied Programming in C; Interface and Digital Electronics; Machine Intelligence; Deep Learning; Digital Integrated Circuit Design; Engineering Analysis; Human Factors in AI.

Obafemi Awolowo University

2019 – 2021

M.Tech., Electronics and Telecommunication Engineering, Distinction

Ile-Ife, Nigeria

Thesis: *Duct Propagation: Aiding Naval Communication at Sea.*

Federal University of Technology Owerri

2002 – 2007

B.Eng., Electrical and Electronic Engineering; GPA: 3.99/5.00

Owerri, Nigeria

Best Female Graduating Student. Final-year project: *Design and Construction of Microcomputer Based Wireless Security System.*

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, Embedded C, Assembly

Embedded Systems: STM32, MSP432, Arm Cortex-M, FreeRTOS, CMSIS, CubeMX, Keil uVision

Engineering: Real-time systems, sensor-based measurement, signal processing, control systems, experimental instrumentation

Software: MATLAB, Simulink, Git, Visual Studio Code, Windows, Linux

ENGINEERING EXPERIENCE

Graduate Research Assistant

Aug 2023 – Present

Respiratory Technologies Laboratory, Rochester Institute of Technology

Rochester, NY

- Contribute to the design, commissioning, calibration, validation, deployment, and performance evaluation of Wireless Personal Use Monitors (wPUM) used to measure puff topography in natural-environment studies.
- Use programmable emissions systems and calibrated reference flow meters to evaluate monitor response and support pressure-to-flow calibration across product-specific monitor configurations.
- Compile and analyze laboratory and field datasets to assess measurement reliability, calibration stability, sensor behavior, and postdeployment performance.
- Support electronic-cigarette emissions testing and product characterization, including analysis of puff flow rate, duration, volume, timing, and aerosol yield.
- Develop analysis workflows in MATLAB and Python and maintain research records, figures, and technical documentation for federally funded studies.

Weapon Electrical Officer, Lieutenant Commander

Apr 2012 – Sep 2023

Nigerian Navy

Nigeria

- Supervised installation, maintenance, troubleshooting, and repair of shipboard electrical, electronic, navigation, communication, control, computer, and weapon systems.
- Led technical personnel responsible for equipment readiness, power distribution, and critical auxiliary systems during sea-going and shore assignments.
- Served in engineering appointments aboard NNS THUNDER and NNS UNITY and at NNS PATHFINDER, Forward Operating Base Bonny, and Naval Headquarters.
- Participated in technical training and acquisition activities for NNS UNITY in China as a member of the commissioning crew and sailed the newly acquired vessel from China to Nigeria.
- Final appointment: Staff Officer II, Naval Engineering Branch, Naval Headquarters, Abuja.

SELECTED ENGINEERING PROJECTS

- **Real-Time and Embedded Systems:** Implemented timing, servo-control, PWM, task-scheduling, and queue-simulation projects using STM32, FreeRTOS, and Embedded C.
- **Interface and Digital Electronics:** Completed UART communication, optical-sensor characterization, motor-control, heart-rate-monitor, operational-amplifier, and filter-design laboratory projects using MSP432 and embedded interfaces.
- **C Programming:** Developed coursework projects involving arrays, file handling, data processing, and algorithm implementation.

PUBLICATIONS, MANUSCRIPTS & PRESENTATIONS

- Mbaka, C. G., Hensel, E. C., and Robinson, R. J. “Assessing Monitors for Characterizing the Natural Environment Puff Topography of Tobacco Product Users.” *Manuscript under review*, 2026.
- Mbaka, C. G., Hensel, E. C., and Robinson, R. J. “Quality Control and Data Integrity Assessment of Ambulatory Puff Topography Data Using Gen 3 wPUM Monitors.” *Manuscript under review*, 2026.
- Mbaka, C. G., Hensel, E. C., and Robinson, R. J. “Effect of Topography Monitoring on User Behaviour Using Evening Salivary Cotinine in Observation Study 8.” *Manuscript under review*, 2026.
- Mbaka, C. G., Hensel, E. C., Kelly, L., Holman, L., Eddingsaas, N. C., and Robinson, R. J. “Ambulatory Puff Topography of Participants in a Weeklong Natural Environment Observation Study.” Poster, AAAR 43rd Annual Conference, 2025.
- Mbaka, C. G., Hensel, E. C., Garzon, C., and Robinson, R. J. “HYDE Brand Electronic Cigarette Aerosol Emissions Characterization.” Poster, Air Pollution and Health Research Symposium, 2024.

TECHNICAL REPORT

Hensel, E. C., Mbaka, C. G., Jayasekera, S., Eddingsaas, N. C., Butler, J. D., and Robinson, R. J. *PTC0001 Laboratory Study Protocol (Final)*. FDA Contract No. 75F40125C00098, Respiratory Technologies Laboratory, Rochester Institute of Technology, 2026. Approved by the FDA technical review panel.

HONORS & PROFESSIONAL SERVICE

- Abstract Reviewer, NIH Tobacco Regulatory Science Meeting, 2026.
- Chief of the Naval Staff Award for Best Overall Graduating Student, Junior Command and Staff Course, 2021.
- Director’s Award for Best Overall Graduating Student, Department of Maritime Warfare, 2021.
- Director’s Award for Leadership, Department of Maritime Warfare, 2021.
- Commandant’s Exemplary Leadership Award, Nigerian Navy Engineering College, 2018.
- Best Female Graduating Student in Electrical/Electronic Engineering, Federal University of Technology Owerri, 2007.

SELECTED TECHNICAL TRAINING

- Nigerian Offshore Patrol Vessel Technical Training, China Shipbuilding Industry Corporation, China.
- Engineering Watchkeeping Certification.
- Officers’ Application Course, Nigerian Navy Engineering College.
- NATO Civil-Military Training, Sweden; Maritime Security Training, Ghana.

PROFESSIONAL AFFILIATIONS

IEEE | IEEE Women in Engineering | IEEE Young Professionals | Nigerian Society of Engineers | Council for the Regulation of Engineering in Nigeria (COREN)